



**BLUESTOCK MUTUAL FUND ANALYTICS
CAPSTONE PROJECT
PROJECT REPORT**

1. Executive Summary

This project focuses on analyzing mutual fund data to provide meaningful insights into fund performance, investor behavior, portfolio concentration, and SIP continuity. The project combines data engineering, exploratory data analysis, financial performance analytics, and interactive dashboarding to create a complete mutual fund analytics solution.

The data was collected from multiple mutual fund datasets including fund master information, NAV history, assets under management, SIP inflows, investor transactions, portfolio holdings, and benchmark indices. The data was cleaned, validated, stored in SQLite, analyzed using Python and SQL, and visualized using Power BI.

The project successfully developed a scalable analytics pipeline that enables investors and analysts to evaluate mutual fund schemes using risk-adjusted metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Value at Risk (VaR), and Conditional Value at Risk (CVaR). The final dashboard provides an intuitive interface for exploring fund performance and investor trends.

The outcomes of the project demonstrate how data analytics can improve investment decision-making and provide actionable insights for mutual fund stakeholders.

- Build an automated ETL pipeline.
- Clean and validate mutual fund datasets.
- Store data in SQLite database.
- Perform exploratory data analysis.
- Calculate performance metrics.
- Build Power BI dashboards.
- Perform advanced analytics.
- Generate investment insights and recommendations.

2. Data Sources & ETL Design

2.1 Data Sources

The project utilizes multiple datasets covering different aspects of mutual fund operations. These datasets provide information related to fund characteristics, investor behavior, portfolio composition, and performance metrics.

Datasets Used:

- Fund Master Dataset
- NAV History Dataset
- AUM by Fund House Dataset
- Monthly SIP Inflows Dataset
- Category Inflows Dataset
- Industry Folio Count Dataset
- Scheme Performance Dataset
- Investor Transactions Dataset
- Portfolio Holdings Dataset
- Benchmark Indices Dataset

2.2 ETL Architecture

The ETL process was designed to automate the movement of mutual fund data from raw datasets into a structured analytics environment.

ETL Workflow:

1. Extract

- Import raw CSV datasets.
- Validate file structure and schema.

2.Transform

- Remove duplicates.
- Handle missing values.
- Standardize column names.
- Validate AMFI codes.
- Correct data types.

3. Load

- Store cleaned datasets in SQLite database.
- Create analytical tables for reporting.

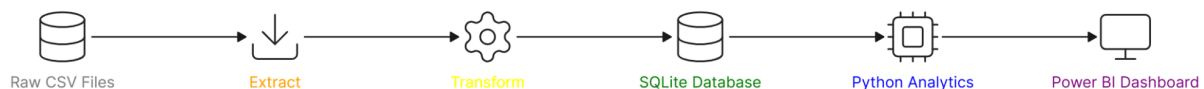
4. Analytics Layer

- Perform EDA and performance analysis.
- Calculate financial metrics.
- Generate advanced analytics outputs.

5. Reporting Layer

- Build Power BI dashboards.
- Generate reports and visualizations.

ETL Architecture Diagram



3. Data Cleaning & Validation

The raw mutual fund datasets required several preprocessing steps before analysis. Data cleaning was performed using Python and Pandas to improve data quality and consistency.

3.1 Data Cleaning

Data Cleaning Activities:

- Removal of duplicate records.
- Handling missing and null values
- Standardization of column names
- Date format conversion.
- Numeric data type correction.
- Consistency checks across datasets.
- Validation of AMFI codes.

These steps ensured that the datasets were accurate and suitable for downstream analytics. Data validation was performed to ensure reliability and integrity of the datasets

3.2 Data Validation .

Validation Checks:

- Verification of unique AMFI codes.
- Detection of duplicate records
- Range validation for numerical fields.
- Validation of date formats
- Consistency checks between related datasets.
- Identification of missing values.

The validation process significantly reduced data quality issues and improved confidence in analytical results.

3.3 Tools Used

- Python
- Pandas
- NumPy
- SQLite
- Jupyter Notebook
- Git & GitHub
- Power BI

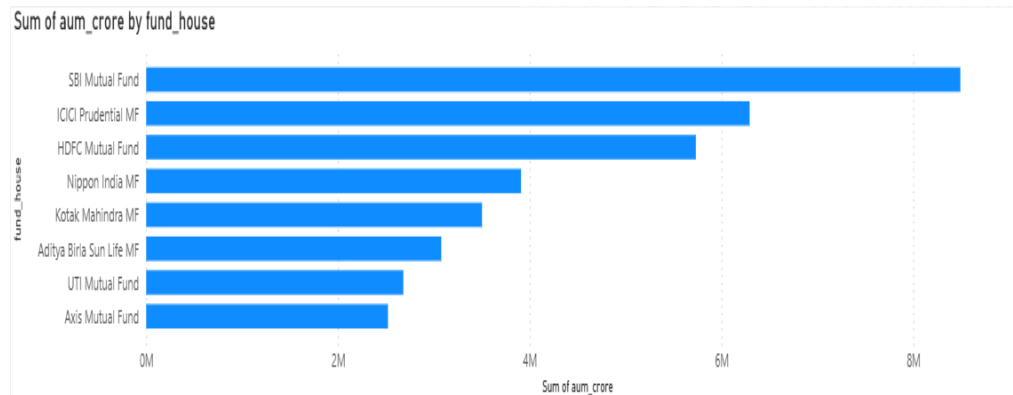
4. Exploratory Data Analysis (EDA)

4.1 Category-wise Fund Analysis

Exploratory Data Analysis was conducted to understand fund distribution, category trends, SIP inflows, and investor participation patterns.

The category-wise analysis revealed that Equity Funds accounted for the largest share of schemes in the dataset, followed by Debt Funds and Hybrid Funds. This indicates a strong investor preference toward growth-oriented investment products.

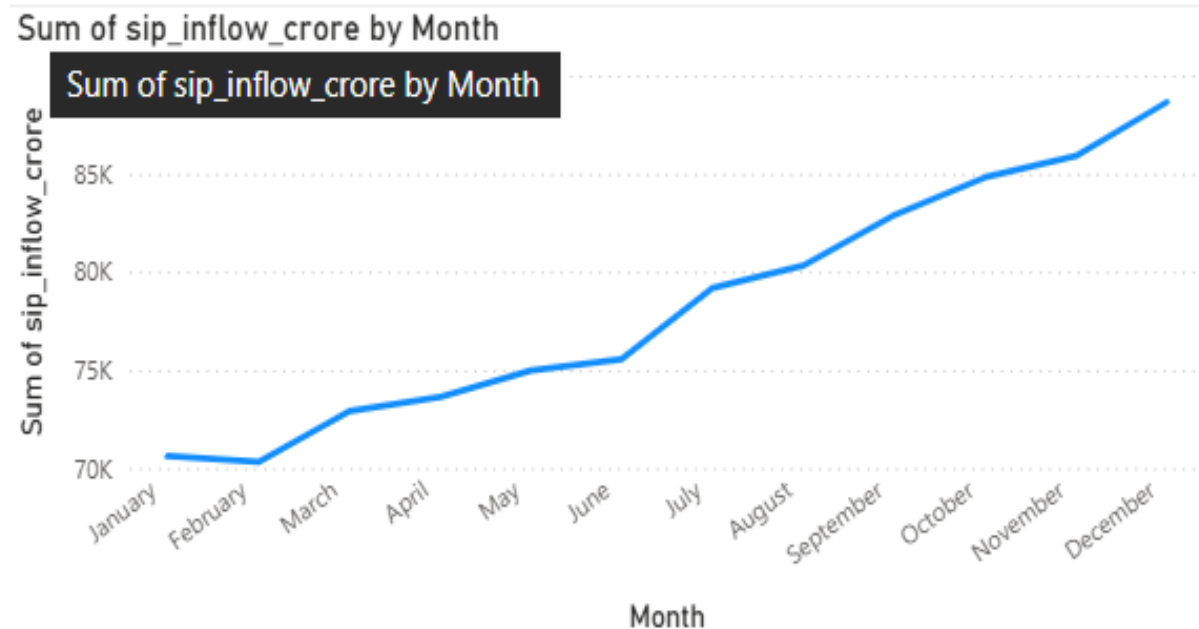
The analysis also highlighted differences in fund size, investor participation, and performance characteristics across categories.



4.2 SIP Inflow Analysis

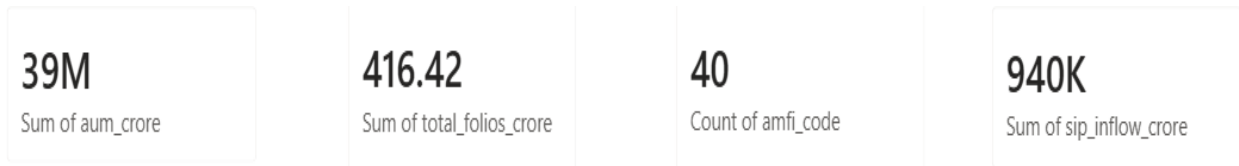
Monthly SIP inflows were analyzed to identify investment trends over time. The analysis showed consistent growth in SIP contributions, reflecting increasing retail investor participation in mutual funds.

Higher SIP inflows indicate greater investor confidence and long-term investment discipline. Seasonal fluctuations were observed, but the overall trend remained positive.



4.3 AUM Analysis

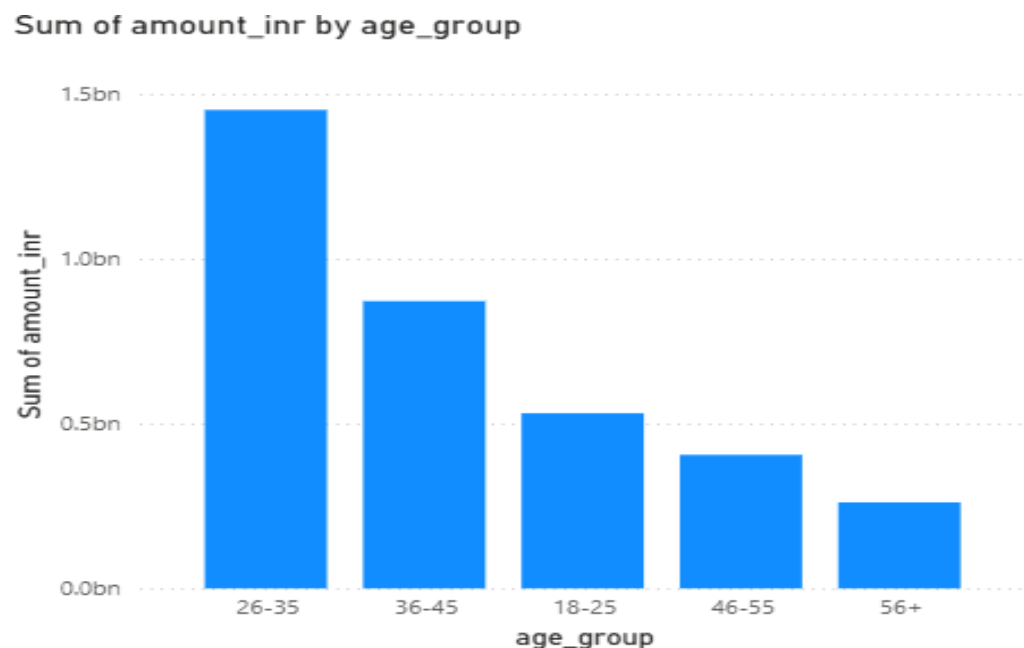
Assets Under Management (AUM) analysis was performed to evaluate the relative size of fund houses and investment products. Large fund houses managed a substantial portion of total industry assets, indicating strong brand recognition and investor trust. AUM distribution also highlighted concentration among leading asset management companies.



4.4 Investor Folio Analysis

Folio analysis provided insights into investor participation levels across mutual fund categories.

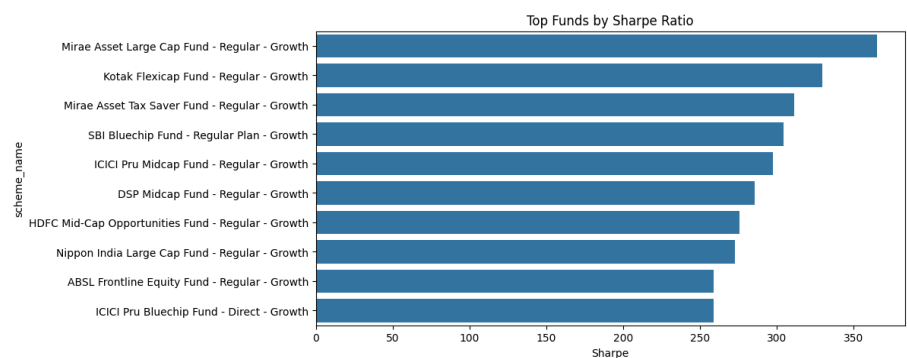
Equity-oriented schemes attracted the highest number of investor folios, demonstrating their popularity among retail investors. The folio count analysis also reflected broader market participation and investor diversification trends.



5. Performance Analytics

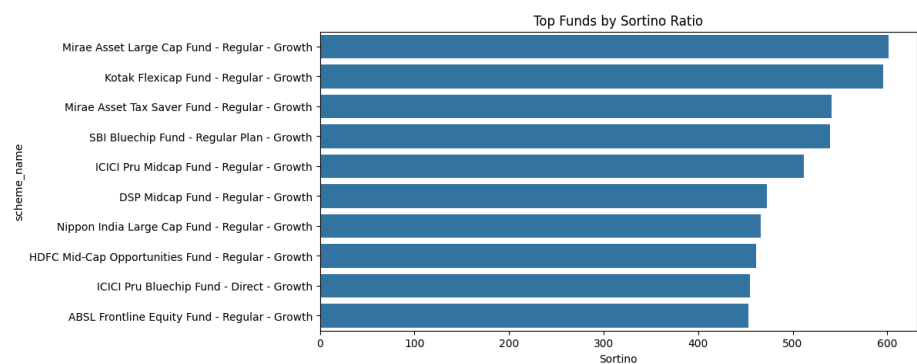
5.1 Sharpe Ratio Analysis

The Sharpe Ratio was used to evaluate risk-adjusted returns across mutual fund schemes. Funds with higher Sharpe Ratios generated better returns per unit of risk taken. The analysis identified several top-performing funds that consistently delivered superior risk-adjusted performance compared to their peers. Investors seeking efficient portfolios can use Sharpe Ratio as an important decision-making metric.



5.2 Sortino Ratio Analysis

The Sortino Ratio measures risk-adjusted returns by focusing only on downside volatility. Unlike the Sharpe Ratio, it penalizes harmful volatility while ignoring positive fluctuations. Funds with higher Sortino Ratios demonstrated stronger downside protection and more stable performance during adverse market conditions.



5.3 Alpha and Beta Analysis

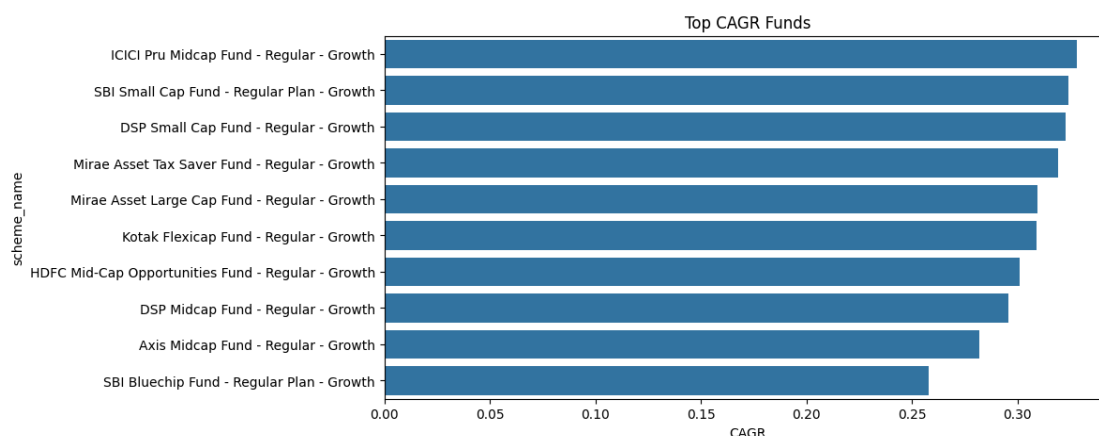
Alpha and Beta metrics were used to assess fund performance relative to benchmark indices. Alpha measures excess return generated by a fund manager, while Beta measures market sensitivity. Funds with positive Alpha values and moderate Beta values were considered attractive because they delivered superior returns without excessive market risk.

	amfi_code	Alpha	Beta \	
0	100016	-0.089699	0.788441	
1	100025	0.033199	0.060713	
2	100033	0.114423	0.991590	
3	101206	0.097443	0.737015	
4	101207	-0.350741	2.878184	
				scheme_name
0				HDFC Top 100 Fund - Regular Plan - Growth
1				HDFC Short Term Debt Fund - Regular - Growth
2				HDFC Mid-Cap Opportunities Fund - Regular - Gr...
3				ABSL Frontline Equity Fund - Regular - Growth
4				ABSL Small Cap Fund - Regular - Growth

5.4 CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated to measure long-term growth performance of mutual fund schemes.

The CAGR analysis helped identify schemes that consistently compounded investor wealth over time and provided a standardized comparison across different investment products.



5.5 Key Performance Findings

Key observations from the performance analysis include:

- Several schemes delivered strong risk-adjusted returns based on Sharpe Ratio.
- Sortino Ratio analysis indicated effective downside risk management among top funds.
- Positive Alpha values highlighted successful active fund management.
- Moderate Beta values indicated balanced market exposure.
- CAGR results demonstrated the long-term wealth creation potential of selected mutual funds.

6. Advanced Analytics

6.1 Value at Risk (VaR) & Conditional VaR

Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated to assess downside investment risk.

VaR estimates the maximum expected loss under normal market conditions at a specified confidence level, while CVaR measures the average loss beyond the VaR threshold. These metrics help investors understand potential downside exposure and identify high-risk schemes.

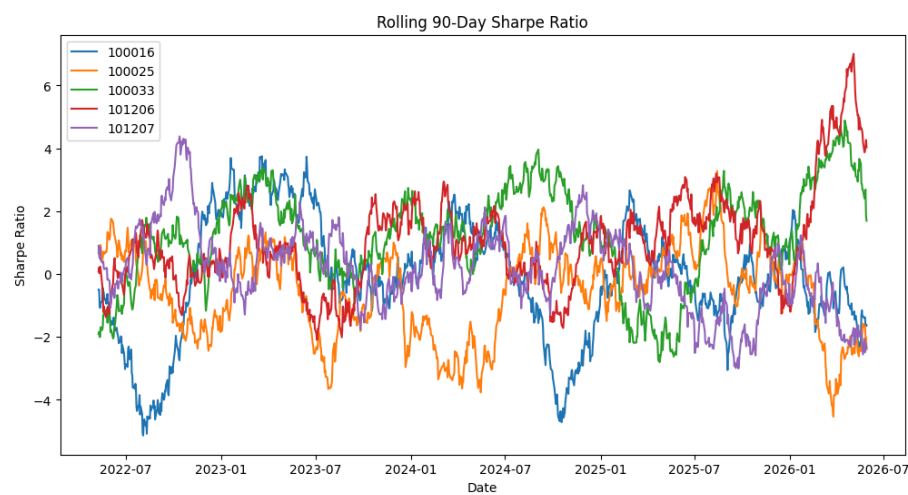
	amfi_code	VaR_95	CVaR_95
22	119599	-0.026859	-0.032384
17	119095	-0.026188	-0.031667
4	101207	-0.026021	-0.032459
11	118634	-0.025438	-0.032304
21	119598	-0.024507	-0.030595
39	149324	-0.023483	-0.031036
7	102886	-0.019220	-0.023251
2	100033	-0.019034	-0.023456
25	120505	-0.018892	-0.024342
16	119094	-0.018480	-0.024260

6.2 Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was calculated to monitor changes in risk-adjusted performance over time.

The rolling analysis revealed periods of strong and weak performance, demonstrating that fund efficiency is dynamic and influenced by changing market conditions.A rolling 90-day Sharpe Ratio was calculated to monitor changes in risk-adjusted performance over time.

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6.3 Investor Cohort Analysis

Investor cohort analysis grouped investors based on their first investment year. This approach helped evaluate investment behavior, average contribution size, and fund preferences across different investor generations.

The analysis showed that newer cohorts contributed larger investment amounts, reflecting increasing awareness and participation in mutual fund investments.

	investor_id	transaction_date	amfi_code	transaction_type	amount_inr	state	city	city_tier	age_group	gender	annual_income_lakh	payment_mode	kyc_status	cohort_year
0	INV003054	2024-01-01	119092	SIP	1834	Telangana	Hyderabad	T30	56+	Female	77.1	UPI	Verified	2024
1	INV002952	2024-01-01	148567	Redemption	392882	Punjab	Amritsar	B30	18-25	Male	7.1	Cheque	Verified	2024
2	INV003420	2024-01-01	118636	SIP	912	Haryana	Faridabad	B30	36-45	Male	47.2	Mandate	Verified	2024
3	INV003436	2024-01-01	118634	SIP	1102	Maharashtra	Mumbai	T30	36-45	Female	54.4	Cheque	Pending	2024
4	INV004691	2024-01-01	119094	Lumpsum	8682	Delhi	Noida	T30	26-35	Male	14.5	Net Banking	Pending	2024

6.4 SIP Continuity Analysis

SIP continuity analysis evaluated investor discipline by measuring the average gap between SIP transactions. Investors with average gaps greater than 35 days were classified as at-risk. The majority of investors maintained regular SIP schedules, indicating strong long-term investment behavior.

	investor_id	transaction_date	amfi_code	transaction_type	amount_inr	state	city	city_tier	age_group	gender	annual_income_lakh	payment_mode	kyc_status	cohort_year	gap_days
19621	INV000001	2024-11-04	120505	SIP	44856	Haryana	Gurugram	T30	36-45	Male	19.9	UPI	Verified	2024	NaN
24448	INV000001	2025-01-19	125497	SIP	3090	Haryana	Gurugram	T30	36-45	Male	19.9	Cheque	Pending	2024	76.0
5650	INV000002	2024-03-29	149322	SIP	2830	Maharashtra	Pune	T30	46-55	Male	24.0	Mandate	Verified	2024	NaN
16803	INV000002	2024-09-21	120841	SIP	2354	Maharashtra	Pune	T30	46-55	Male	24.0	Mandate	Verified	2024	176.0
31881	INV000002	2025-05-17	119094	SIP	2690	Maharashtra	Pune	T30	46-55	Male	24.0	Cheque	Verified	2024	238.0

6.5 Portfolio Concentration (HHI)

The Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration. Higher HHI values indicate that a fund is concentrated in fewer holdings, increasing concentration risk. Lower HHI values suggest greater diversification and lower portfolio-specific risk.

	amfi_code	HHI
11	119092	0.206448
3	101207	0.200700
18	119599	0.174751
4	102885	0.174709
7	118632	0.168298
29	148568	0.167930
21	120505	0.157570
22	120506	0.153794
27	125498	0.152414
23	120841	0.149680

	amfi_code	HHI
count	34.000000	34.000000
mean	121457.058824	0.140262
std	14988.778388	0.024946
min	100016.000000	0.107349
25%	118633.250000	0.122955
50%	119575.000000	0.136496
75%	120842.750000	0.151731
max	149324.000000	0.206448

6.6 Fund Recommendation System

A simple recommendation engine was developed to suggest suitable funds based on investor risk appetite. The recommender classifies investors into Low, Moderate, and High risk categories and recommends top-performing schemes based on Sharpe Ratio rankings within each category.

	scheme_name	risk_category	Sharpe_Ratio
34	Mirae Asset Large Cap Fund - Regular - Growth	Moderate	1.448291
22	Kotak Flexicap Fund - Regular - Growth	Moderately High	1.306744
0	SBI Bluechip Fund - Regular Plan - Growth	Moderate	1.208267

6.7 Advanced Analytics Findings

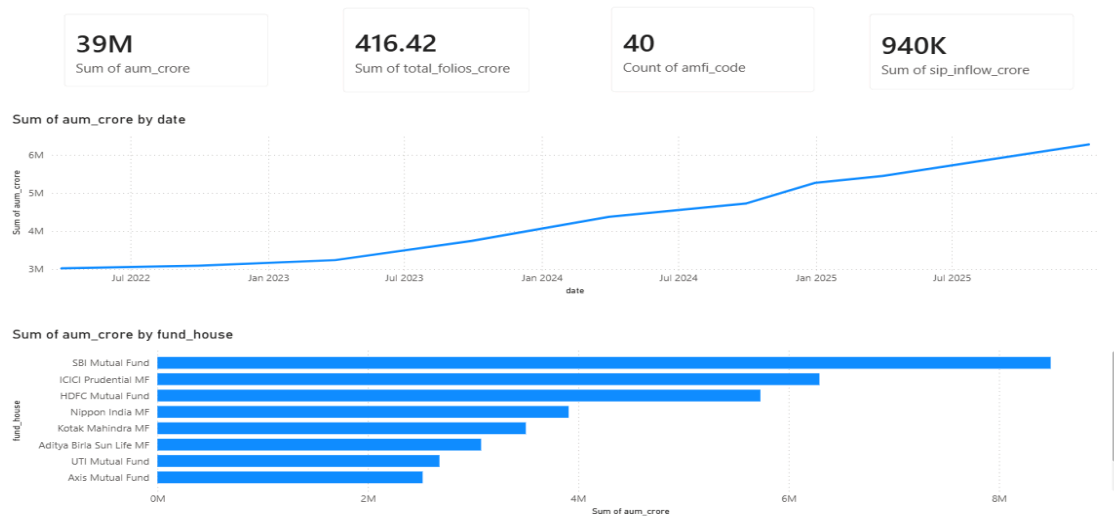
Key findings from advanced analytics include:

- Certain funds exhibited significantly higher VaR and CVaR values, indicating elevated downside risk.
- Rolling Sharpe Ratios varied considerably across market cycles.
- Recent investor cohorts invested larger amounts than earlier cohorts.
- SIP continuity remained strong among most investors.
- Some equity funds displayed high portfolio concentration based on HHI analysis.
- The recommendation engine successfully identified suitable funds for different risk profiles.

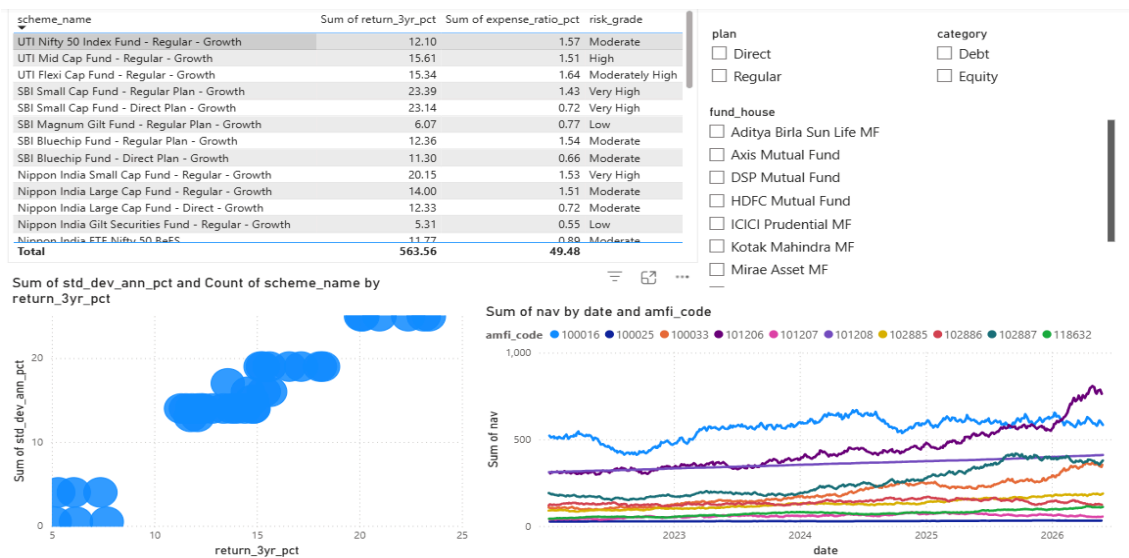
7. Power BI Dashboard Overview

An interactive Power BI dashboard was developed to visualize mutual fund performance, investor behavior, SIP trends, and portfolio analytics. The dashboard enables users to explore fund performance through interactive filters, drill-through functionality, KPI cards, and advanced visualizations. The dashboard consists of four primary pages designed for different analytical objectives.

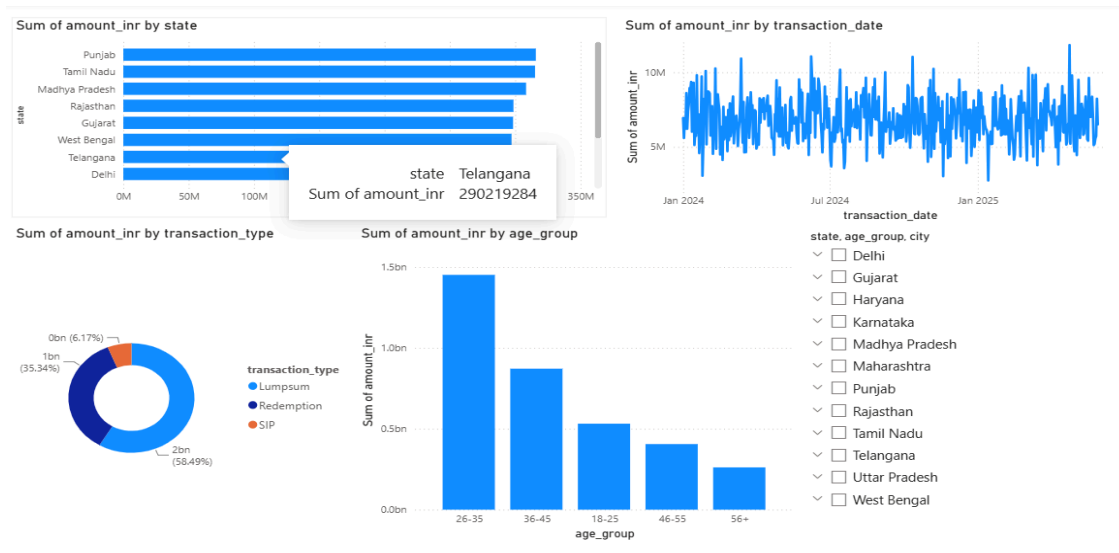
7.1 Industry Overview Dashboard



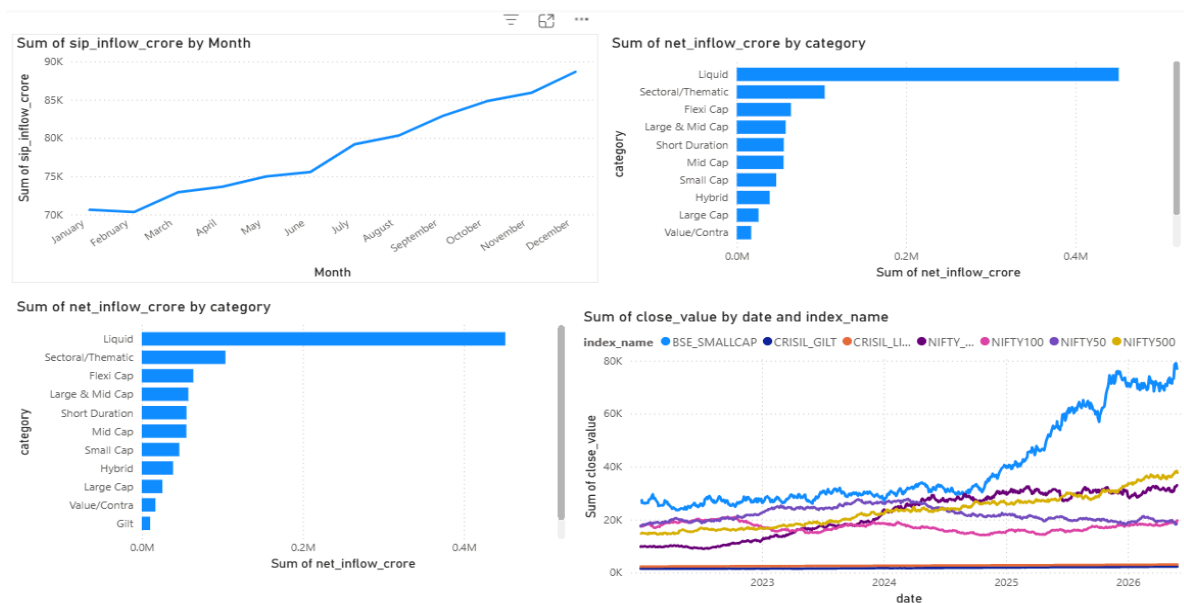
7.2 Fund Performance Dashboard



7.3 Investor Analytics



7.4 SIP & Market Trends Dashboard



8. Limitations

Although the project successfully demonstrates a complete mutual fund analytics workflow, several limitations exist.

- The project uses static datasets rather than real-time market data.
- Historical performance may not predict future performance.
- Macroeconomic variables were not incorporated into the analysis.
- The recommendation system uses rule-based logic rather than machine learning.
- Data quality depends on source accuracy and completeness.

9. Recommendations

Future enhancements can significantly improve the analytical capabilities of the platform.

- Integrate real-time NAV and market data APIs.
- Develop machine learning-based fund recommendation models.
- Add predictive analytics for future fund performance.
- Deploy dashboards to cloud platforms.
- Incorporate macroeconomic and sector-level indicators.
- Implement automated reporting and alert systems.

10. Conclusion

This project successfully developed a comprehensive mutual fund analytics platform using Python, SQL, SQLite, Power BI, and advanced financial analytics techniques.

The solution demonstrated the complete data analytics lifecycle including data ingestion, cleaning, storage, analysis, visualization, and reporting. Through performance metrics, advanced risk analytics, investor behavior analysis, and dashboarding, the project generated meaningful investment insights.

The final deliverables provide a scalable framework that can be extended to support real-world investment analysis and decision-making processes.

11. References

- [AMFI India](#)
- [NSE India](#)
- [Mutual Fund Industry Reports](#)
- [Python Documentation](#)
- [Pandas Documentation](#)
- [SQLite Documentation](#)
- [Microsoft Power BI Documentation](#)
- [Bluestock Internship Guidelines](#)